

Submit your Jupyter notebook with the answers to gradescope by 3:40pm.

1. Assuming the parameter a is larger than 2 and b is larger than a in the integral

$$\int_a^b \frac{1}{x-2} dx,$$

compute its value.

2. Consider $f_k(x_1, x_2, \dots, x_n) = x_k(x_{1+(k-2) \bmod n} + x_{1+k \bmod n}) + 2x_k - 3$.

Define a function F which takes on input a list X of variables and an index k in the range of X . $F(X, k)$ returns $f_k(x_1, x_2, \dots, x_n)$. Show that your function works for $n = 8$.